GridGuard: Enterprise Wildfire Intelligence Network

Business Requirements Document (v1.5 - Refined)

Executive Summary

GridGuard is an advanced wildfire intelligence platform designed to protect high-value forest assets and public safety through a distributed mesh of autonomous sensors. By reducing the "Time-to-Detection" from hours to under 60 seconds, GridGuard empowers the Department of Forestry & Conservation (DFC) to transition from **reactive fire suppression** to **proactive incident containment**, significantly reducing environmental loss and operational costs.

The Opportunity: Closing the Detection Gap

Current wildfire detection relies on satellite imagery (limited by refresh rates and cloud cover) and human reporting (subjective and delayed).

- **The Cost of Delay:** Every minute of delay in wildfire detection increases suppression costs exponentially and elevates the risk to life and property.
- **The Solution:** A field-level, "always-on" intelligence layer that identifies incipient fires (<1 acre) with high-confidence consensus, ensuring that emergency resources are deployed only when and where they are truly needed.

Strategic Business Objectives

1. Operational Efficiency & ROI

- **Accelerated Response:** Maintain a sub-60-second alert latency to ensure containment while fires are small and manageable.
- **Resource Optimization:** Utilize sensor consensus to virtually eliminate "false positive" deployments, saving millions in unnecessary aviation and personnel costs.

2. Scalable Infrastructure

- **Massive Deployment:** Architected to support up to 50,000 nodes per region, providing comprehensive coverage for vast, remote territories.
- **Sustainability:** Low-power, solar-optimized hardware ensures a multi-year lifecycle with minimal maintenance overhead.

3. Business Continuity

- **Resilient Connectivity:** Redundant communication paths (LoRa-to-Satellite) ensure that critical alerts reach dispatch even when local cellular or power grids fail.

Jobs-To-Be-Done (JTBD): Outcome-Driven Logic

GridGuard is "hired" to achieve specific, measurable improvements in the wildfire management lifecycle. We define success by the system's ability to:

1. Detection & Verification

- **Minimize** the time between an ignition event (<1 acre) and a verified notification reaching a tactical responder.

- **Maximize** the probability of identifying a fire in remote or densely canopied areas where satellite visibility is obstructed.
- **Minimize** the frequency of "False Positive" alerts by leveraging multi-sensor consensus before escalation.

2. Network Proliferation & Installation

- **Minimize** the time required to achieve a "fully commissioned and communicating" status for a sensor node during physical installation.
- **Maximize** the success rate of "first-trip" installations in environments with zero cellular or satellite connectivity.
- **Minimize** the manual data entry and technical training required for field personnel to expand the sensor grid.

3. Operational Coordination & Dispatch

- **Maximize** the signal-to-noise ratio of incoming emergency data to prevent responder alert fatigue.
- **Minimize** the likelihood of deploying high-cost suppression resources (aviation/crews) to non-fire events or "ghost" heat signatures.
- **Maximize** the consistency of data formatting for legacy national emergency dispatch systems (NEDS).

4. System Trust & Security

- **Minimize** the risk of system compromise or data spoofing by unauthorized edge devices.
- **Maximize** the integrity and non-repudiation of every ingested alert signal through the network.
- **Minimize** the operational impact of hardware failures through automated heartbeat and health reporting.

Key Stakeholder Impact

Forest Rangers (Strategic Response)

- **Value Prop:** Real-time situational awareness.
- **Outcome:** Rangers receive verified, high-confidence alerts on mobile devices, allowing for immediate tactical decisions without waiting for aerial confirmation.

Field Technicians (Operational Deployment)

- **Value Prop:** Rapid, frictionless installation.
- **Outcome:** "Zero-touch" provisioning via mobile QR scanning allows technicians to expand the network 50% faster than traditional IoT deployments, even in offline environments.

Dispatch & Operations (Risk Management)

- **Value Prop:** Alert integrity and system reliability.
- **Outcome:** Integration with National Emergency Dispatch Systems (NEDS) provides a filtered, prioritized feed of events, preventing "alert fatigue" during high-risk seasons.

Core Capabilities & Deliverables

- **Intelligence Layer (Cloud Platform):** A centralized "Brain" that aggregates sensor data, applies consensus logic, and manages the global device registry.
- **Connectivity Gateway (Range Hubs):** Edge computing units that buffer data and bridge remote forest locations to the global satellite network.
- **Field Command App:** A ruggedized mobile interface for network health monitoring and rapid device provisioning.
- **Enterprise Integration:** Seamless data hand-off to legacy NEDS systems via robust, queued API architecture.

Architectural Commitments

These architectural decisions are embedded into the system design to ensure reliability and trust:

1. **Tiered Alert Engine:** Alerts are categorized into *Warning*, *Confirmed*, and *Exception* tiers. This determines whether an event is simply displayed on a dashboard or triggers an immediate national dispatch.
2. **Cloud-Centric Decision Logic:** While sensors detect and hubs buffer, the Cloud remains the single source of truth for consensus and alert generation to maintain data integrity.
3. **Offline-First Provisioning:** The mobile app supports local caching and eventual synchronization, allowing network expansion in areas with zero connectivity.
4. **Buffered & Graceful Dispatch Integration:** To prevent overloading legacy NEDS systems, outgoing messages are queued and throttled. During extreme events, the system automatically switches to "summary mode" to prevent message storms.
5. **Device Authentication at Ingest:** Identity verification happens at both the edge and cloud layers to reject spoofed or unregistered devices before data is ever processed.

Key Performance Indicators (KPIs)

Objective	Target Metric
Detection Speed	≥ 90% of verified fires detected in < 60 seconds
Alert Accuracy	< 1% False Positive rate (Consensus-verified)
Network Uptime	99.9% Cloud availability
Operational Ease	< 5 minutes to provision a new node in the field
Data Integrity	< 2% alert loss during legacy system downtime